Individual Study Plan

Adaptive Strategy Learning in Multi-Agent Imperfect-Information Environments

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1 Individual Study Plan

1.1 Research on the possibilities for applying Artificial Intelligence in computer games.

Superscript numerals (e.g. [15]) refer to entries in the Glossary at the end of this document.

1.2 1. Introduction

1.2.1 1.1 Research Context and Significance

Multi-agent systems operating under conditions of imperfect informationgl[5] represent one of the most challenging and practically relevant areas of artificial intelligence research. In such systems, autonomous agents must make strategic decisions without complete knowledge of the environment state or the intentions of other participants — a setting that arises naturally in cybersecurity (adversarial network defense), financial markets (competing algorithmic traders), autonomous system coordination, and fraud detection.

The mathematical framework for reasoning about such interactions is provided by extensive-form gamesgl[4] with imperfect information, a branch of game theorygl that models sequential decision-making when players hold private information. Over the past decade, substantial progress has been achieved in computing equilibrium strategiesgl[7] for large-scale imperfect-information games. Landmark systems include Libratus (Brown and Sandholm, 2017) and Pluribus (Brown and Sandholm, 2019), which defeated professional human players in two-player and six-player poker respectively, as well as more general architectures such as ReBeL (Brown et al., 2020) and Student of Games (Schmid et al., 2023), which unify game-solving approaches across perfect- and imperfect-information domains.

Despite these advances, a critical limitation persists: state-of-the-art systems compute fixed equilibrium strategies that do not adapt to the specific behavioral patterns exhibited by encountered opponents. In practice, real-world adversaries rarely play optimally, and the ability to detect and exploit systematic deviations from equilibrium — while maintaining guarantees against being exploited in return — constitutes a fundamental open problem. This problem, known as safe opponent exploitationgl[3], has been investigated primarily in two-player zero-sumgl[8] settings. Its extension to multiplayer environments, where coalitions may form and dissolve dynamically, remains largely unexplored in the literature.

The present study plan is organized as a progressive program of fifteen study steps, grouped into seven thematic phases:

- **Foundation** (Chapters 1–2): Reinforcement learninggl and game-theoretic fundamentals required by all subsequent work.
- **Scaling the Toolbox** (Chapters 3–4): Monte Carlo CFRgl variants and game abstraction techniques for handling larger game instances.
- **Neural Methods for Games** (Chapters 5–6): Neural network approximations of equilibrium strategies and end-to-end game AI architectures.
- **Opponent Modeling and Exploitation** (Chapters 7–8): Inference from behavioral tracesgl and safe exploitation algorithms — the thesis-critical core.
- **Multi-Agent Dynamics** (Chapters 9–11): Multi-agent reinforcement learninggl, population-based traininggl, and coalition dynamicsgl in competitive settings.
- **Data-Driven Approaches** (Chapters 12–13): Sequence models and behavioral analysis pipelines connecting theory to real-world data.
- **Integration** (Chapters 14–15): Evaluation framework construction and research frontier mapping.

Throughout the study plan, Kuhn Pokergl and Leduc Hold'emgl serve as the primary implementation testbeds. These games are chosen deliberately as pedagogical vehicles: their small state spaces (12 and approximately 936 information sets, respectively) and known analytical equilibria permit exact verification of every algorithm, while their imperfect-information structure retains the strategic complexity that the thesis demands. By working within well-understood domains, each step concentrates on algorithmic concepts rather than domain-specific engineering, maximizing the volume of theoretical material covered within the allotted timeframe. In later phases, the study plan validates generality beyond poker by applying the developed methods to matrix games, Goofspiel, So Long Suckergl, and anonymized real-world behavioral data.

The planned thesis contributions are formulated in domain-agnostic terms — none is specific to poker or to any other single game. The Behavioral Adaptation Framework (Contribution 1) targets arbitrary imperfect-information games; the Multi-Agent Safe Exploitation heuristics (Contribution 2) will be defined over general N-player extensive-form structures; and the Evaluation Methodology (Contribution 3) is intended for cross-domain applicability.

The study plan is aligned with the milestones of the university individual plan. Upon completion of each thematic phase, the corresponding material will be incorporated into Chapter I of the dissertation — covering the state-of-the-art analysis and the formulation of relevance, objectives, tasks, and thesis statements. In this way, the literature review and foundational exposition accumulate incrementally rather than being composed retrospectively. Chapter 15 produces a detailed Chapter I outline and publication pipeline.

1.2.2 1.2 Research Objective and Expected Contributions

The primary objective of this doctoral research is to develop an adaptive AI agent to be used in various computer games. The agent will start from a safe strategy but will be able to improve its outcome against sub-optimal opponents, thus removing the limitations of previous systems which rarely dynamically adapt their behavior. The research program is structured around three planned contributions:

Contribution 1 — Behavioral Adaptation Framework. The goal is to develop a general method for inferring and adapting to opponent strategies from observed action sequences in real time, applicable to arbitrary imperfect-information games. This contribution aims to address the gap between static equilibrium computation and dynamic, opponent-aware play.

Contribution 2 — Multi-Agent Safe Exploitation. This contribution will explore tractable heuristic approaches — including KL-regularized exploitationgl (constraining the exploitative policy to remain close to a safe reference strategy via a Kullback–Leibler divergence penalty) and equal-share baselines — for safe exploitation in small N-player games (three-player Kuhn and Leduc poker variants), seeking to extend existing results from the two-player zero-sum case. The scope is limited to the empirical validation of practical heuristics rather than a general safety theorem for arbitrary N-player settings.

Contribution 3 — Evaluation Methodology. The aim is to design a domain-agnostic framework for measuring agent adaptabilitygl and robustnessgl across different game environments and opponent populations, with emphasis on identifying failure modes of existing evaluation approaches and demonstrating where standard metrics may provide misleading assessments.

1.3 2. Phase A — Foundation (Chapters 1–2)

1.3.1 2.1 Phase Overview

This phase covers the foundational material required for all subsequent work: reinforcement learning basics (value estimation, policy optimization, experience replay) and game-theoretic fundamentals (extensive-form games, Nash equilibrium computation, and exploitability measurement). A solid command of both domains is essential before proceeding to the more specialized topics of later phases.

1.3.2 2.2 Chapter 1 — Reinforcement Learning Basics

Contribution Alignment. The reinforcement learning methods studied in this chapter — value estimation, policy gradients, and experience replay — will recur throughout the thesis, particularly in the opponent modeling and adaptive exploitation components of the planned contributions.

Literature.

1. Sutton, R.S. and Barto, A.G. (2018). *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press.
2. Mnih, V., Kavukcuoglu, K., Silver, D. et al. (2015). "Human-level control through deep reinforcement learning." *Nature*, 518(7540), pp. 529–533.
3. Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O. (2017). "Proximal Policy Optimization Algorithms." Preprint.

Practical Tasks.

- Implement a Deep Q-Network[16] (DQN) agent with experience replay[9] and target network; train on a discrete control task (CartPole-v1).
- Implement a Proximal Policy Optimization[17] (PPO) agent with generalized advantage estimation; train on a continuous control task (LunarLander-v3).
- Compare learning curves of both agents; analyze sensitivity to key hyperparameters (learning rate, batch size, discount factor).
- Validate both implementations against established reference results.

1.3.3 2.3 Chapter 2 — Game Theory and Counterfactual Regret Minimization

Contribution Alignment. The extensive-form game representation and CFR algorithm introduced here will provide the formal framework and baseline equilibrium computation used throughout the thesis. Exploitability[3], defined and implemented in this chapter, will serve as the primary quantitative metric across all three contributions.

Literature.

1. Shoham, Y. and Leyton-Brown, K. (2008). *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.* Cambridge University Press.
2. Neller, T.W. and Lanctot, M. (2013). "An Introduction to Counterfactual Regret Minimization." Technical report, Gettysburg College.
3. Zinkevich, M., Johanson, M., Bowling, M. and Piccione, C. (2007). "Regret Minimization in Games with Incomplete Information." *Advances in Neural Information Processing Systems 20 (NeurIPS).*

Practical Tasks.

- Construct a complete game engine for Kuhn Poker[19] (three-card deck, two players, one betting round), including game tree enumeration and terminal payoff computation.
- Implement vanilla CFR[15] with regret[18] accumulation and average strategy computation; verify convergence to the known analytical Nash equilibrium[7] of Kuhn Poker.
- Implement best response[1] computation and exploitability[3] measurement; verify $O(1/\sqrt{T})$ convergence rate on a log-log scale.
- Cross-validate the resulting strategy profile against the OpenSpiel reference implementation.

1.4 3. Phase B — Scaling the Toolbox (Chapters 3–4)

1.4.1 3.1 Phase Overview

Phase A established a working CFR[15] solver on the minimal Kuhn Poker[19] benchmark, but vanilla CFR requires a full traversal of the game tree on every iteration — an approach that becomes infeasible as games grow. This phase will introduce two complementary scaling mechanisms: Monte Carlo sampling methods that reduce per-iteration cost, and game abstraction techniques that reduce the game tree itself. These tools are needed to bridge the gap between toy benchmarks and the medium-scale games on which later thesis work will be developed.

1.4.2 3.2 Chapter 3 — CFR Variants and Monte Carlo Methods

Contribution Alignment. Monte Carlo CFR variants will provide the computationally tractable equilibrium computation needed for medium-scale games, which will underpin the empirical work in later contributions. CFR+ accelerates convergence, enabling equilibrium computation for games beyond the reach of vanilla CFR. Understanding the variance characteristics of different sampling schemes will inform later design decisions in opponent modeling.

Literature.

1. Lanctot, M., Waugh, K., Zinkevich, M. and Bowling, M. (2009). "Monte Carlo Sampling

for Regret Minimization in Extensive Games." *Advances in Neural Information Processing Systems 22 (NeurIPS).*

2. Tammelin, O., Burch, N., Johanson, M. and Bowling, M. (2015). "Solving Heads-Up Limit Texas Hold'em." *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI).*
3. Chen, B. and Ankenman, J. (2006). *The Mathematics of Poker.* ConJelCo. Chapters 1–8.

Practical Tasks.

- Construct a Leduc Hold'em[20] game engine (six-card deck, two rounds, one community card, ~936 information sets[6]).
- Implement external-sampling MCCFR on both Kuhn and Leduc; verify convergence to the same Nash equilibrium as vanilla CFR with lower wall-clock time.
- Implement CFR+ with regret flooring, alternating updates, and linear averaging; demonstrate approximately ten-fold convergence speedup over vanilla CFR on Kuhn Poker.
- Compare convergence profiles of all variants: exploitability[3] versus iterations and versus wall time.

1.4.3 3.3 Chapter 4 — Game Abstraction and Scaling Imperfect-Information Games

Contribution Alignment. This chapter will study how game abstraction — both lossless and lossy — affects the quality of computed equilibria, and how subgame solving[22] can refine strategies during play. These techniques are directly relevant to later work on safe exploitation (where abstraction quality bounds exploitation risk) and opponent modeling (where abstraction granularity determines the ability to distinguish opponent types).

Literature.

1. Gilpin, A. and Sandholm, T. (2007). "Lossless Abstraction of Imperfect Information Games." *Journal of the ACM*, 54(5).
2. Johanson, N., Burch, N., Valenzano, R. and Bowling, M. (2013). "Evaluating State-Space Abstractions in Extensive-Form Games." *Proceedings of the 12th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS).*
3. Kroer, C. and Sandholm, T. (2016). "Imperfect-Recall Abstractions with Bounds in Games." Preprint.
4. Brown, N. and Sandholm, T. (2017). "Safe and Nested Subgame Solving for Imperfect-Information Games." Preprint.

Practical Tasks.

- Implement lossless abstraction via suit isomorphism detection on Leduc Hold'em; verify that the resulting Nash equilibrium[7] is identical to the unabstracted solution.

- Implement lossy card bucketing with configurable granularity; measure exploitability degradation as the number of buckets decreases.
- Construct an Extended Leduc variant (four ranks, two suits, multiple bet sizes, ~5000+ information sets) as a scaling testbed.
- Build a combined abstraction pipeline composing card bucketing, action abstraction, and suit isomorphism; evaluate on a Pareto frontier of abstraction size versus exploitability gap.

1.5 4. Phase C — Neural Methods for Games (Chapters 5–6)

1.5.1 4.1 Phase Overview

Phases A and B established tabular equilibrium solvers and abstraction techniques for medium-scale games. However, tabular methods store explicit strategy and regret values at every information set[6] — an approach whose memory requirements grow linearly with game size and become prohibitive for large-scale domains. This phase replaces tabular storage with neural network function approximation, enabling equilibrium computation without explicit game tree enumeration. Chapter 5 introduces the core approximation methods (Deep CFR[25], NFSP[27]), while Chapter 6 studies the complete game-solving architectures (DeepStack, Libratus, Pluribus, ReBeL[28], Student of Games) that integrate these components into competition-grade systems. Together, these chapters complete the algorithmic toolbox from which the thesis contributions are developed: Contribution 1 builds on the public belief state[29] framework of ReBeL, Contribution 2 addresses the theoretical gap exposed by Pluribus's multiplayer success without formal safety guarantees, and Contribution 3 draws on the exploitability metrics applied uniformly across all five architectures.

1.5.2 4.2 Chapter 5 — Neural Equilibrium Approximation (Deep CFR, DREAM)

Contribution Alignment. Deep CFR[25] and NFSP[27] will be studied as methods for computing equilibrium strategies in games too large for tabular solvers — a capability needed for the planned behavioral adaptation work. NFSP's anticipatory parameter, which interpolates between equilibrium and exploitative play, foreshadows the exploitation–safety tradeoff central to the thesis. The information state tensor encoding developed here will serve as the standard input representation for later neural architectures.

Literature.

1. Brown, N., Lerer, A., Gross, S. and Sandholm, T. (2019). "Deep Counterfactual Regret Minimization." *Proceedings of the 36th International Conference on Machine Learning (ICML).*
2. Steinberger, E. (2019). "Single Deep Counterfactual Regret Minimization." *Proceedings of the 34th AAAI Conference on Artificial Intelligence (2020).*
3. Heinrich, J. and Silver, D. (2016). "Deep Reinforcement Learning from Self-Play in Imperfect-Information Games." Preprint.

Practical Tasks.

- Develop a reusable information state tensor encoding for Leduc Hold'em[20], designed for compatibility with all subsequent neural implementations.
- Implement Deep CFR with advantage networks, reservoir sampling, and strategy network distillation; verify that predicted counterfactual values[2] match tabular baselines on Kuhn Poker[19].
- Implement NFSP as a comparative baseline; evaluate the effect of the anticipatory parameter on the Nash–exploitation tradeoff.
- Compare convergence behavior (exploitability[3] versus wall time) of Deep CFR, tabular MCCFR[23], and NFSP on Leduc Hold'em.

1.5.3 4.3 Chapter 6 — End-to-End Game AI Architectures

Contribution Alignment. This chapter will survey five landmark game-solving systems that define the current state of the art. ReBeL's public belief state[29] framework is of particular relevance to the planned belief-based opponent modeling (Contribution 1). Pluribus demonstrates empirical success in multiplayer poker without formal safety guarantees — highlighting the theoretical gap that Contribution 2 will seek to address. The exploitability metric applied uniformly across all architectures will inform the evaluation methodology design (Contribution 3).

Literature.

1. Moravcik, M., Schmid, M., Burch, N., Lisý, V., Morrill, D., Bard, N., Davis, T., Waugh, K., Johanson, M. and Bowling, M. (2017). "DeepStack: Expert-Level Artificial Intelligence in Heads-Up No-Limit Poker." *Science*, 356(6337), pp. 508–513.
2. Brown, N. and Sandholm, T. (2018). "Superhuman AI for Heads-Up No-Limit Poker: Libratus Beats Top Professionals." *Science*, 359(6374), pp. 418–424.
3. Brown, N. and Sandholm, T. (2019). "Superhuman AI for Multiplayer Poker." *Science*, 365(6456), pp. 885–890.
4. Brown, N., Bakhtin, A., Lerer, A. and Hu, Q. (2020). "Combining Deep Reinforcement Learning and Search for Imperfect-Information Games." *Advances in Neural Information Processing Systems (NeurIPS).*
5. Schmid, M., Moravcik, M., Burch, N. et al. (2023). "Student of Games: A Unified Learning Algorithm for Both Perfect and Imperfect Information Games." *Science Advances*, 9(46).
6. Brown, N. and Sandholm, T. (2018). "Depth-Limited Solving for Imperfect-Information Games." *Advances in Neural Information Processing Systems (NeurIPS).*

Practical Tasks.

- Implement a simplified ReBeL system ("ReBeL-Lite") for Leduc Hold'em: public belief state representation with Bayesian updates, local PBS-CFR solver, and a value network trained via self-play.

- Verify that training iterations produce monotonically decreasing exploitability on Leduc.
- Construct a comparative analysis of all five architectures (DeepStack, Libratus, Pluribus, ReBeL, Student of Games), mapping shared components (MCCFR[23], subgame solving[22], neural value estimation, depth-limited search) and identifying the evolutionary path from single-player to multiplayer to unified systems.

1.6 5. Phase D — Opponent Modeling and Exploitation (Chapters 7–8)

1.6.1 5.1 Phase Overview

This phase addresses the central research question: how should an agent adapt its play to a specific opponent? The preceding phases built the algorithmic toolbox — equilibrium solvers, abstraction, and neural approximation — but did not yet tackle opponent-aware play. Chapter 7 will introduce inference mechanisms that convert observed action sequences into beliefs about opponent behavior. Chapter 8 will cover algorithms that translate those beliefs into profitable yet safe strategy adjustments. Together, these chapters will form the foundation for Contribution 1 (Behavioral Adaptation) and expose the theoretical limitations that Contribution 2 (Multi-Agent Safe Exploitation) will need to address.

1.6.2 5.2 Chapter 7 — Opponent Modeling — Inference from Behavioral Traces

Contribution Alignment. Bayesian opponent modeling[31] will serve as the inference component of the planned Behavioral Adaptation Framework (Contribution 1). Three modeling paradigms will be studied — type-based models, continuous parametric models, and consistent convergent estimators — each offering different tradeoffs between assumptions, convergence speed, and robustness. The multiplayer extension, which requires joint beliefs over all opponents, will expose computational challenges relevant to Contribution 2. Non-stationarity handling will address an open problem relevant to the evaluation methodology (Contribution 3).

Literature.

1. Southey, F., Bowling, M., Larson, B., Piccione, C., Burch, N., Billings, D. and Rayner, C. (2005). "Bayes' Bluff: Opponent Modelling in Poker." *Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI).*
2. Bard, N., Johanson, M., Burch, N. and Bowling, M. (2013). "Online Implicit Agent Modelling." *Proceedings of the 12th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS).*
3. Ganzfried, S. and Sun, Q. (2016). "Bayesian Opponent Exploitation in Imperfect-Information Games." Preprint.
4. Ganzfried, S., Wang, K.A. and Chiswick, M. (2022). "Opponent Modeling in Multiplayer

Imperfect-Information Games." Preprint.
5. Ganzfried, S. (2025). "Consistent Opponent Modeling in Imperfect-Information Games." Preprint.
6. Shoham, Y. and Leyton-Brown, K. (2008). *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.* Cambridge University Press. Chapter 7: Learning and Teaching.

Practical Tasks.

- Construct an opponent type library with five or more distinct strategies (tight-aggressive, loose-passive, Nash equilibrium, exploitative, random) for both Kuhn Poker[19] and Leduc Hold'em[20].
- Implement an observation buffer handling partial observability (showdown versus non-showdown information extraction).
- Implement a type-based Bayesian opponent model[31] with Dirichlet-multinomial posterior inference and a continuous Bayesian model with per-information-set action distribution estimation.
- Implement a consistent opponent modeler following the Ganzfried (2025) sequence-form projected gradient descent approach.
- Build an adaptive exploitation pipeline integrating opponent modeling and best-response[1] computation.
- Conduct a head-to-head model comparison measuring convergence speed, final exploitation rate, and robustness to unknown opponent types.
- Execute non-stationarity tests demonstrating model behavior under opponent type switching.

1.6.3 5.3 Chapter 8 — Safe Exploitation — Theory, Algorithms, and Real-Time Search

Contribution Alignment. This chapter covers the exploitation–safety tradeoff[33], which is the theoretical core of the thesis. Key topics include the Restricted Nash Response (RNR[32]), the Safety Theorem of Ganzfried and Sandholm (2015) — which provides formal guarantees in two-player zero-sum settings but fails in N-player games, exposing the gap that Contribution 2 will address — subgame exploitation methods for real-time safe play (relevant to Contribution 1), and teaching attack[34] resilience testing as a prototype for the adversarial evaluation methodology of Contribution 3.

Literature.

1. Johanson, M., Bowling, M. and Zinkevich, M. (2007). "Computing Robust Counter-Strategies." *Advances in Neural Information Processing Systems (NeurIPS).*
2. Ganzfried, S. and Sandholm, T. (2015). "Safe Opponent Exploitation." *ACM Transactions on Economics and Computation*, 3(2).
3. Liu, W., Wang, H., Guo, T. and Xing, J. (2022). "Safe Opponent-Exploitation Subgame

Refinement." *Advances in Neural Information Processing Systems (NeurIPS).*

4. Jeary, J. and Turrini, P. (2023). "Safe Opponent Exploitation For Epsilon Equilibrium Strategies." Preprint.
5. Ge, C., Zhu, Y. et al. (2024). "Safe and Robust Subgame Exploitation in Imperfect Information Games." *Proceedings of the 41st International Conference on Machine Learning (ICML).*
6. Milec, D., Kovařík, V. and Lisý, V. (2025). "Adapting Beyond the Depth Limit: Counter Strategies in Large Imperfect Information Games." Preprint.
7. Shoham, Y. and Leyton-Brown, K. (2008). *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.* Cambridge University Press. Sections 3.4 and 4.6.

Practical Tasks.

- Implement a safety checker module computing worst-case exploitability[3] and verifying safety constraints against the blueprint strategy.
- Implement an exploitation metrics module measuring exploitation value and safety violations.
- Build a Restricted Nash Response (RNR[32]) solver with configurable safety parameter $p \in [0, 1]$.
- Implement the Ganzfried safe exploitation solver enforcing the Safety Theorem guarantee.
- Implement a prime-safe exploitation extension handling ε-equilibrium baselines (connecting abstraction quality from Chapter 4).
- Build an SES-style subgame exploitation solver with gadget construction for real-time safe exploitation.
- Implement an adaptation safety[35] checker following the Ge et al. (2024) safety notion.
- Generate exploitation–safety Pareto frontier plots for all methods across multiple opponent types.
- Integrate the full pipeline: Chapter 7 opponent models[30] feeding Chapter 8 exploitation engine, evaluated end-to-end.
- Conduct a teaching attack[34] stress test with deceptive opponent switching behavior mid-game and robustness analysis.

1.7 6. Phase E — Multi-Agent Dynamics (Chapters 9–11)

1.7.1 6.1 Phase Overview

The preceding phases focused on two-player imperfect-information games. Phase E will transition the study to multi-agent settings, where new challenges arise: non-stationarity[36] from simultaneously learning agents, credit assignment in joint-reward environments, and the emergence of coalitions[41]. Chapter 9 introduces multi-agent RL paradigms (CTDE[37], PSRO[38]), Chapter 10 scales to population-based training[39] and evolutionary game theory, and Chapter 11 applies these tools to coalition formation in free-for-all games — crystallizing the theoretical gap at the core of Contribution 2.

1.7.2 6.2 Chapter 9 — Multi-Agent RL — Coordination, Competition, and Communication

Contribution Alignment. This chapter will provide the algorithmic vocabulary for extending the thesis from two-player to multi-agent settings. The CTDE[37] paradigm introduces the architectural pattern — centralized training, decentralized execution — used throughout the remainder of the thesis. PSRO[38] provides a population-based framework relevant to defining safety in multi-agent populations (Contribution 2). LOLA[40] contributes the insight of modeling an opponent's learning dynamics, extending the Bayesian modeling of Chapter 7 from static inference to dynamic anticipation (Contribution 1).

Literature.

1. Lowe, R., Wu, Y., Tamar, A., Harb, J., Abbeel, P. and Mordatch, I. (2017). "Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments." *Advances in Neural Information Processing Systems (NeurIPS).*
2. Rashid, T., Samvelyan, M., de Witt, C.S., Farquhar, G., Foerster, J. and Whiteson, S. (2018). "QMIX: Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning." *Proceedings of the International Conference on Machine Learning (ICML).*
3. Yu, C., Velu, A., Vinitsky, E., Gao, J., Wang, Y., Baez, A. and Fishi, S. (2022). "The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games." *Advances in Neural Information Processing Systems (NeurIPS).*
4. Foerster, J., Chen, R.Y., Al-Shedivat, M., Whiteson, S., Abbeel, P. and Mordatch, I. (2018). "Learning with Opponent-Learning Awareness." *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS).*
5. Lanctot, M., Zambaldi, V., Gruslys, A., Lazaridou, A., Tuyls, K., Pérolat, J., Silver, D. and Graepel, T. (2017). "A Unified Game-Theoretic Approach to Multiagent Reinforcement Learning." *Advances in Neural Information Processing Systems (NeurIPS).*
6. Sukhbaatar, S., Szlam, A. and Fergus, R. (2016). "Learning Multiagent Communication with Backpropagation." *Advances in Neural Information Processing Systems (NeurIPS).*

Practical Tasks.

- Construct a matrix game testbed (Prisoner's Dilemma, Matching Pennies, Stag Hunt, Battle of the Sexes) with verified Nash equilibria.
- Implement independent PPO learners as a baseline demonstrating non-stationarity[36] failures in multi-agent settings.
- Implement MADDPG (centralized critic, decentralized actors) and MAPPO (PPO with centralized value function) as CTDE[37] representatives.
- Implement PSRO[38] with population-based meta-Nash computation and RL best-response oracle; verify convergence to Nash equilibrium on Kuhn Poker[19] and Leduc Hold'em[20].

- Integrate a CommNet differentiable communication channel with MADDPG; evaluate the benefit of emergent communication on cooperative tasks.
- Produce a comparative evaluation table: independent learning versus CTDE versus PSRO across all test environments and Goofspiel.

1.7.3 6.3 Chapter 10 — Population-Based Training and Evolutionary Game Theory

Contribution Alignment. Population-based training and the AlphaStar league architecture will be studied as examples of implicit opponent modeling at population scale, complementing the explicit Bayesian modeling of Chapter 7 (Contribution 1). The spinning top decomposition[42] — distinguishing genuine skill improvement from non-transitive cycling — will be adopted into the evaluation methodology (Contribution 3). EGTA[43] will provide meta-Nash analysis extending exploitability assessment to population settings.

Literature.

1. Jaderberg, M., Dalibard, V., Osindero, S., Czarnecki, W.M. et al. (2017). "Population Based Training of Neural Networks." Preprint.
2. Jaderberg, M., Czarnecki, W.M., Dunning, I. et al. (2019). "Human-Level Performance in First-Person Multiplayer Games with Population-Based Deep Reinforcement Learning." *Science*, 364(6443), pp. 859–865.
3. Vinyals, O., Babuschkin, I., Czarnecki, W.M. et al. (2019). "Grandmaster Level in StarCraft II Using Multi-Agent Reinforcement Learning." *Nature*, 575(7782), pp. 350–354.
4. Balduzzi, D., Garnelo, M., Bachrach, Y., Czarnecki, W.M., Pérolat, J., Jaderberg, M. and Graepel, T. (2019). "Open-Ended Learning in Symmetric Zero-Sum Games." *Proceedings of the International Conference on Learning Representations (ICLR).*
5. Tuyls, K., Pérolat, J., Lanctot, M. et al. (2018). "A Generalised Method for Empirical Game Theoretic Analysis." *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS).*
6. Hofbauer, J. and Sigmund, K. (2003). "Evolutionary Game Dynamics." *Bulletin of the American Mathematical Society*, 40(4), pp. 479–519.

Practical Tasks.

- Implement a replicator dynamics simulator on matrix games; verify convergence to Nash equilibrium and evolutionary stable strategies on Prisoner's Dilemma, Hawk-Dove, and Stag Hunt, and verify cycling on Rock-Paper-Scissors. Generate phase portraits.
- Implement the spinning top decomposition[42] (Balduzzi et al., 2019); apply to PSRO[38] meta-game payoff matrices from Chapter 9 and to the league meta-game from this chapter. Compute the transitive ratio as a diagnostic metric.
- Build a PBT league for Leduc Hold'em[20] with three agent roles (main agents, main exploiters,

league exploiters), prioritized matchmaking, periodic agent freezing, and PBT explore-exploit population updates.

- Conduct EGTA[43] analysis: construct the empirical normal-form game over the league population and compute meta-Nash equilibrium; verify that the meta-Nash mixture exploitability does not exceed that of the best individual agent.
- Produce a comparative evaluation: league versus PSRO (Chapter 9) versus single self-play agent versus MCCFR[23] Nash strategy (Chapter 3), measuring exploitability, Elo rating, effective population diversity, and strategy clustering.

1.7.4 6.4 Chapter 11 — Dynamic Coalition Formation in Competitive Free-For-All Games

Contribution Alignment. This chapter crystallizes the central theoretical gap of the thesis. In two-player games, safe exploitation uses Nash equilibrium as the safety baseline (Chapter 8). In N-player free-for-all games, Nash equilibrium is both computationally intractable and strategically insufficient — it ignores coalition structures. The piKL regularization approach of Bakhtin et al. (2022) suggests replacing equilibrium-based safety with behavioral-prior-based safety, a shift that Contribution 2 will seek to formalize. Coalition detection will extend opponent modeling from individual player types to multi-agent social structure (Contribution 1). Shapley-value[44] credit decomposition combined with EGTA[43] will provide the basis for N-player evaluation methodology (Contribution 3).

Literature.

1. Sharan, M. and Adak, C. (2024). "Reinforcing Competitive Multi-Agents for Playing 'So Long Sucker'." Preprint.
2. De Carufel, J.-L. and Jerade, M.R. (2024). "So Long Sucker: Endgame Analysis." Preprint.
3. Bakhtin, A., Wu, D.J., Lerer, A., Gray, J., Jacob, A.P., Farina, G., Miller, A.H. and Brown, N. (2022). "Mastering the Game of No-Press Diplomacy via Human-Regularized Reinforcement Learning and Planning." Preprint.
4. Chalkiadakis, G., Elkind, E. and Wooldridge, M. (2011). *Computational Aspects of Cooperative Game Theory.* Morgan and Claypool.
5. Wang, J., Zhang, Y., Kim, T.-K. and Gu, Y. (2020). "Shapley Q-value: A Local Reward Approach to Solve Global Reward Games." *Proceedings of the 34th AAAI Conference on Artificial Intelligence.*

Practical Tasks.

- Build a verified So Long Sucker[45] environment with coalition tracking, rich state representation, and correctness validation against the endgame analysis of De Carufel and Jerade (2024).

- Implement a coalition detection module that infers implicit alliances from observed chip-placement behavior using help/harm matrices, extending the opponent modeling methodology of Chapter 7 to multi-agent alliance inference.
- Adapt Shapley Q-value[44] decomposition to the competitive free-for-all setting, distributing each action's credit among all players according to marginal coalition contributions.
- Train four-player MAPPO agents with Shapley-decomposed rewards via self-play; compare against a sparse-reward baseline replicating Sharan and Adak (2024).
- Apply EGTA[43] (Chapter 10) to construct the four-player payoff tensor and compute meta-Nash over the agent population; apply the spinning top decomposition[42] to quantify the non-transitive structure of coalition dynamics.
- Produce a comparative evaluation: coalition-aware agents versus sparse-reward agents versus random baselines, measuring win rate, coalition formation frequency, Shapley variation, and game length.

1.8 7. Phase F — Data-Driven Approaches (Chapters 12–13)

1.8.1 7.1 Phase Overview

The preceding phases developed methods entirely within synthetic, self-play environments. Phase F will bridge the gap between theoretical methods and real-world behavioral data. Chapter 12 introduces sequence models (Decision Transformers[46]) and assesses large language model agents in strategic settings. Chapter 13 applies the resulting pipeline to anonymized real-world poker hand histories, constructing player embeddings[48], behavioral classification systems, and a collusion detection[50] module.

1.8.2 7.2 Chapter 12 — Sequence Models and LLM Agents in Strategic Settings

Contribution Alignment. This chapter will study the Decision Transformer[46] architecture and its adversarially robust variant (ARDT[47]), which recovers near-Nash strategies from offline data — a potential alternative path for Contribution 2 that bypasses intractable equilibrium computation in N-player settings. A critical limitation of return conditioning in stochastic environments will constrain the design of the real-world data pipeline in Chapter 13. The multi-paradigm comparison table produced here will serve as a prototype for the evaluation framework of Contribution 3.

Literature.

1. Chen, L., Lu, K., Rajeswaran, A., Lee, K., Grover, A., Laskin, M., Abbeel, P., Srinivas, A. and Mordatch, I. (2021). "Decision Transformer: Reinforcement Learning via Sequence Modeling." *Advances in Neural Information Processing Systems (NeurIPS).*
2. Paster, K., McIlraith, S. and Ba, J. (2022). "You Can't Count on Luck: Why Decision Transformers and RvS Fail in Stochastic Environments." Preprint.

3. Tang, X., Marques, A., Kamalaruban, P. and Bogunovic, I. (2024). "Adversarially Robust Decision Transformer." *Advances in Neural Information Processing Systems (NeurIPS).*
4. Janner, M., Li, Q. and Levine, S. (2021). "Offline Reinforcement Learning as One Big Sequence Modeling Problem." *Advances in Neural Information Processing Systems (NeurIPS).*
5. Guertler, L., Cheng, B., Yu, S., Liu, B., Choshen, L. and Tan, C. (2025). "TextArena." Preprint.
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Practical Tasks.

- Generate a poker trajectory dataset (50,000+ Kuhn Poker[19] hands) using CFR agents from Chapter 3; design the poker state tensor encoding that will transfer to the real-world data pipeline in Chapter 13.
- Adapt and train a Decision Transformer[46] for poker state and action spaces; validate return-to-go conditioning and demonstrate the stochasticity limitation (Paster et al., 2022) whereby the model conflates lucky card deals with skilled play.
- Implement ARDT[47] with minimax expectile regression; train on mixed-quality opponent data and verify recovery of near-Nash strategies on Kuhn Poker (measured by exploitability[3] against the known analytical equilibrium from Chapter 2).
- Deploy LLM agents on TextArena strategic games to assess capabilities in negotiation, deception, and theory-of-mind reasoning.
- Evaluate LLM agents on Kuhn Poker with standardized metrics: exploitability, bluff frequency, value bet frequency, illegal move rate, and opponent adaptation.
- Produce a unified multi-paradigm comparison table (CFR, Decision Transformer, ARDT, behavioral cloning, LLM agents) on Kuhn Poker with standardized metrics.

1.8.3 7.3 Chapter 13 — Behavioral Analysis Pipelines and Real-World Data

Contribution Alignment. This chapter will apply the behavioral adaptation methodology from Chapter 7, 8, and 12 to anonymized industry data, providing practical validation for Contribution 1. The behavioral deviation from equilibrium play measured on real player data will quantify exploitation opportunities relevant to Contribution 2. The collusion detection[50] module — transferring coalition detection principles from Chapter 11 to a practical fraud detection application — will represent a direct contribution to the evaluation methodology (Contribution 3).

Literature.

1. Wang et al. (2024). "player2vec: A Language Modeling Approach to Understand Player Behavior in Games." Preprint.
2. Kumar, A., Hong, J., Singh, A. and Levine, S. (2022). "When Should We Prefer Offline Rein-

forcement Learning Over Behavioral Cloning?" *Proceedings of the International Conference on Learning Representations (ICLR).*

3. DeLong and Bhatt (2020). "Towards Collusion Detection in Poker."
4. Yan and Browne (2016). "Collusion Detection in Online Poker."
5. Southey, F., Bowling, M., Larson, B., Piccione, C., Burch, N., Billings, D. and Rayner, C. (2005). "Bayes' Bluff: Opponent Modelling in Poker." *Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI).*

Practical Tasks.

- Build a data pipeline from anonymized Playtech hand histories (~50,000–100,000 hands, cash game six-max tables) to structured game records and state tensors; validate for impossible states (duplicate cards, negative stacks, pot mismatches).
- Extend Chapter 12's poker state tensor encoding from Kuhn and Leduc to full Hold'em (52-dimensional one-hot card encoding, position, pot/stack ratios, betting history, street indicator, active player count).
- Compute standard behavioral statistics per player: VPIP[49], PFR, Aggression Factor, WTSD, W$SD, 3-bet percentage, and continuation bet percentage.
- Train player embeddings[48] (player2vec approach) by treating poker actions as tokens and training a Transformer encoder with masked action prediction; extract per-player embedding vectors.
- Apply k-means clustering (four classic archetypes: tight-aggressive, loose-aggressive, tight-passive, loose-passive) and DBSCAN on the embedding space; validate against manual VPIP×PFR classification and test temporal stability.
- Implement a collusion detection[50] module with three signals — co-occurrence anomaly, chip dumping score, and soft play score — combined via weighted composite scoring; validate by injecting synthetic collusion patterns and measuring detection recall (target >90%) and false positive rate (target <5%).
- Apply the Decision Transformer[46] from Chapter 12 to Playtech data; compare return-conditioned versus EV-conditioned training per the Paster et al. stochasticity warning.

1.9 8. Phase G — Integration (Chapters 14–15)

1.9.1 8.1 Phase Overview

Phase G will synthesize the preceding work into two integrative deliverables. Chapter 14 constructs a unified evaluation framework validated across all game types encountered in the study plan — constituting the core of Contribution 3 (Evaluation Methodology). Chapter 15 maps the research frontier, designs the experimental program, produces a Chapter I outline for the dissertation (due November 2026), and establishes the publication pipeline.

1.9.2 8.2 Chapter 14 — Evaluation Frameworks and Exploitability Metrics

Contribution Alignment. This chapter will constitute the core of Contribution 3 directly. The planned three-layer evaluation framework will integrate exploitability computation, population-level ranking (Elo, α-Rank[52], VasE[53]), and statistical confidence quantification (AIVAT[54] variance reduction). It will be validated across Kuhn Poker[19], Leduc Hold'em[20], So Long Sucker[45], and real-world data. For Contribution 1, the framework will measure whether behavioral adaptation produces measurable improvements. For Contribution 2, marginal exploitability[55] will test whether two-player safe exploitation guarantees generalize to multi-agent settings.

Literature.

1. Timbers, F., Bard, N., Lockhart, E., Lanctot, M., Schmid, M., Burch, N., Schrittwieser, J., Hubert, T. and Bowling, M. (2022). "Approximate Exploitability: Learning a Best Response in Large Games." Preprint.
2. Lanctot, M., Larson, K., Bachrach, Y., Marris, L., Li, Z., Bhoopchand, A., Anthony, T., Tanner, B. and Koop, A. (2025). "Evaluating Agents using Social Choice Theory." Preprint.
3. Rowland, M., Omidshafiei, S., Tuyls, K., Perolat, J., Valko, M., Piliouras, G. and Munos, R. (2019). "Multiagent Evaluation under Incomplete Information." Preprint.
4. Burch, N., Johanson, M. and Bowling, M. (2019). "AIVAT: A New Variance Reduction Technique for Agent Evaluation in Imperfect Information Games." *Proceedings of the 33rd AAAI Conference on Artificial Intelligence.*
5. Omidshafiei, S., Papadimitriou, C., Piliouras, G., Tuyls, K. et al. (2019). "α-Rank: Multi-Agent Evaluation by Evolution." *Nature Scientific Reports.*

Practical Tasks.

- Audit and unify all evaluation code from prior steps (exploitability from Chapters 3 and 8, EGTA[43] and spinning top decomposition[42] from Chapter 10, SLS metrics from Chapter 11, behavioral metrics from Chapter 13) into a three-layer evaluation API.
- Construct a bot zoo of reference agents for Kuhn and Leduc covering four tiers: trivial (random, always-call, always-fold), heuristic (tight-aggressive, loose-aggressive, tight-passive), computed (Nash/CFR, DQN), and advanced (PSRO[38], Decision Transformer[46]).
- Implement α-Rank[52] with Fermi selection function; compute stationary distributions via eigenvalue decomposition and analyze sensitivity to the selection pressure parameter.
- Implement VasE[53] with tournament matrix construction from pairwise comparisons across games; compute maximal lotteries via linear programming and intransitive cycle detection.
- Implement AIVAT[54] variance reduction using a CFR-derived control variate with counterfactual adjustment per chance node; verify variance reduction of at least five-fold on Kuhn and ten-fold on Leduc relative to raw evaluation.
- Conduct cross-game validation: full three-layer evaluation on Kuhn, Leduc, So Long Sucker[45]

(marginal exploitability[55]), and Playtech data (AIVAT-adjusted confidence intervals for style classification and collusion detection[50]).

- Produce a disagreement analysis comparing Elo, α-Rank, and VasE rankings, with the spinning top decomposition explaining cases of divergence between transitive and cyclic competitive structures.

1.9.3 8.3 Chapter 15 — Research Frontier Mapping and Contribution Design

Contribution Alignment. This chapter will complete the learning phase and design the research phase. The deliverables will include a research frontier map for each contribution, formal contribution design documents, experimental specifications, a Chapter I outline (25–30 pages, due November 2026), and a publication pipeline through to defense.

Literature.

1. Southey, F., Bowling, M., Larson, B., Piccione, C., Burch, N., Billings, D. and Rayner, C. (2005). "Bayes' Bluff: Opponent Modelling in Poker." *Proceedings of the 21st Conference on Uncertainty in Artificial Intelligence (UAI).*
2. Ganzfried, S. and Sandholm, T. (2015). "Safe Opponent Exploitation." *ACM Transactions on Economics and Computation*, 3(2).
3. Ge, C., Zhu, Y. et al. (2024). "Safe and Robust Subgame Exploitation in Imperfect Information Games." *Proceedings of the 41st International Conference on Machine Learning (ICML).*
4. Ge, C., Wang, Y., Li, W. and Jin, C. (2024). "Securing Equal Share: A Principled Approach for Learning Multiplayer Symmetric Games." Preprint.
5. Milec, D., Kovařík, V. and Lisý, V. (2025). "Adapting Beyond the Depth Limit: Counter Strategies in Large Imperfect Information Games." Preprint.
6. Omidshafiei, S., Papadimitriou, C., Piliouras, G., Tuyls, K. et al. (2019). "α-Rank: Multi-Agent Evaluation by Evolution." *Nature Scientific Reports.*

Practical Tasks.

- Produce a research frontier map documenting, for each of the three contributions, the current state of the art, the identified gap, supporting evidence from the study plan, and a feasibility assessment.
- Validate each identified gap against recent publications (2024–2026), conference proceedings, and existing dissertations; record findings in a gap validation log.
- Write contribution design documents (three documents, two to three pages each) specifying the problem statement, prior art, gap, proposed method, experimental protocol, target publications, timeline, and risk analysis.
- Specify four experiments: (1) behavioral adaptation on Kuhn Poker (player embedding[48] classification accuracy within 50 hands), (2) N-player safe exploitation on three-player Kuhn

(piKL-regularized exploitation versus equal-share baseline), (3) coalition-aware safe exploitation on So Long Sucker[45], and (4) cross-game evaluation framework validation across Kuhn, Leduc, and SLS.

- Produce a Chapter I outline (seven sections, 25–30 pages) covering introduction, foundations, opponent modeling and behavioral adaptation, safe exploitation, evaluation of multi-agent game AI, proposed contributions, and research plan — aligned with the November 2026 deadline.
- Establish the publication pipeline: six target publications mapped to four dissertation chapters with venue selections and submission deadlines through April 2029.

1.10 Glossary

1.10.1 Game Theory

[1] **Best response.** A strategy that maximizes a player's expected payoff given fixed strategies of all other players. Computing a best response against a candidate strategy is the standard method for measuring that strategy's exploitability.

[2] **Counterfactual value.** The expected payoff a player would receive at a given decision point, computed under the assumption that the player acts to reach that point while all other players follow their current strategies. The central quantity in the CFR algorithm.

[3] **Exploitability.** A quantitative measure of how far a strategy profile deviates from Nash equilibrium, defined as the sum of the payoff gains achievable by best-response opponents. An exploitability of zero indicates an exact Nash equilibrium.

[4] **Extensive-form game.** A representation of a sequential game as a tree structure, specifying players, available actions, information sets, chance events, and terminal payoffs. The standard formal model for imperfect-information games.

[5] **Imperfect information.** A property of games in which players do not observe all actions previously taken by other players or by nature. Not to be confused with incomplete information, which refers to uncertainty about the game's rules or other players' preferences.

[6] **Information set.** A set of game states that are indistinguishable to the acting player due to hidden information. A strategy must prescribe the same action distribution across all states within a single information set.

[7] **Nash equilibrium.** A strategy profile in which no player can unilaterally increase their expected payoff by changing their own strategy. In two-player zero-sum games, Nash equilibrium strategies are unexploitable by definition.

[8] **Zero-sum game.** A game in which the payoffs of all players sum to zero at every terminal state. The strictly competitive setting in which regret-minimization algorithms such as CFR are guaranteed to converge to Nash equilibrium.

1.10.2 Reinforcement Learning

[9] Experience replay. A technique in which past transitions (state, action, reward, next state) are stored in a buffer and sampled in mini-batches for training, reducing temporal correlation and improving learning stability.

[10] Markov decision process (MDP). A mathematical model for sequential decision-making in which the transition to the next state depends only on the current state and action. The standard formalism for single-agent reinforcement learning problems.

[11] Policy. A mapping from observations to a probability distribution over actions. Policies may be deterministic (selecting a single action) or stochastic (assigning probabilities to multiple actions).

[12] Policy gradient. A family of reinforcement learning methods that directly optimize a parameterized policy by estimating the gradient of expected cumulative reward with respect to the policy parameters.

[13] Reinforcement learning (RL). A branch of machine learning in which an agent learns to select actions within an environment so as to maximize cumulative reward over time, through a process of trial, error, and delayed feedback.

[14] Value function. A function estimating the expected cumulative reward from a given state (state-value function, V) or state–action pair (action-value function, Q) under a given policy.

1.10.3 Algorithms

[15] Counterfactual Regret Minimization (CFR). An iterative algorithm for approximating Nash equilibrium strategies in two-player zero-sum extensive-form games. Each iteration traverses the game tree, computes counterfactual values, accumulates regrets per action, and updates the strategy via regret matching. The average strategy converges to equilibrium at a rate of $O(1/\sqrt{T})$.

[16] Deep Q-Network (DQN). A reinforcement learning algorithm that approximates the optimal action-value function using a deep neural network, stabilized by experience replay and a periodically updated target network.

[17] Proximal Policy Optimization (PPO). A policy gradient algorithm that stabilizes training by constraining each update step via a clipped surrogate objective function, preventing excessively large policy changes.

[18] Regret matching. A decision rule in which each action is selected with probability proportional to its accumulated positive regret. When applied iteratively within the CFR framework, the resulting average strategy profile converges to Nash equilibrium.

[19] Kuhn Poker. A simplified poker variant with a three-card deck (Jack, Queen, King), two players, and a single betting round. Kuhn Poker has a known analytical Nash equilibrium (game value $\approx -1/18$ for the first player) and serves as the standard minimal testbed for imperfect-information game algorithms.

[20] Leduc Hold'em. A two-round poker variant with a six-card deck (three ranks, two suits) and one community card, producing approximately 936 information sets. Used as an intermediate benchmark between Kuhn Poker and full-scale poker games.

[21] Game abstraction (lossless / lossy). A technique for reducing the size of a game by merging states that are strategically equivalent (lossless) or approximately similar (lossy). Lossless abstractions preserve the exact equilibrium of the original game; lossy abstractions trade solution quality for computational tractability, quantified by the exploitability gap between the abstracted and original solutions.

[22] Subgame solving. A real-time refinement technique that re-solves a portion of the game tree at a finer granularity than the pre-computed blueprint strategy. Safe subgame solving guarantees that the refined strategy is no more exploitable than the original blueprint.

[23] Monte Carlo CFR (MCCFR). A family of CFR variants that sample portions of the game tree rather than traversing it in full on each iteration. Reduces per-iteration cost at the expense of introducing sampling variance. Common instantiations include external sampling (samples opponent and chance actions) and outcome sampling (samples a single trajectory).

[24] CFR+. An accelerated variant of CFR that applies three modifications: flooring negative regrets to zero, alternating which player's strategy is updated, and weighting later iterations more heavily. Empirically achieves $O(1/T)$ convergence — substantially faster than the $O(1/\sqrt{T})$ rate of vanilla CFR.

[25] Deep CFR. A neural variant of CFR that replaces tabular regret and strategy storage with deep neural networks (advantage networks and a strategy network). Training data is generated via external-sampling MCCFR traversals and maintained using reservoir sampling, enabling equilibrium approximation in games too large for tabular methods.

[26] DREAM (Deep Regret minimization with Advantage baselines and Model-free learning). An outcome-sampling variant of Deep CFR that uses baseline subtraction to reduce sampling variance. Achieves comparable solution quality with a single-network architecture.

[27] Neural Fictitious Self-Play (NFSP). An equilibrium-finding algorithm that combines a deep Q-network for best-response computation with a supervised-learning network for average strategy approximation. An anticipatory parameter η interpolates between Nash equilibrium play and exploitative best-response play.

[28] ReBeL (Recursive Belief-based Learning). A game-solving framework that combines self-play reinforcement learning with search over public belief states. Enables AlphaZero-style learning and planning for imperfect-information games without requiring pre-computed blueprint strategies or explicit game abstraction.

[29] Public belief state (PBS). A probability distribution over all possible private information assignments, maintained and updated via Bayes' rule as public actions are observed. The PBS serves as a sufficient statistic for decision-making in imperfect-information games and is the central representation in the ReBeL architecture.

1.10.4 Opponent Modeling and Exploitation

[30] Opponent modeling. The process of inferring an opponent's strategy, intentions, or type from observed behavioral traces during play. Opponent models may be explicit (maintaining a belief distribution over opponent strategies) or implicit (adapting one's own strategy directly from observations without constructing an explicit belief).

[31] Bayesian opponent modeling. An opponent modeling approach that maintains a prior distribution over possible opponent types or action frequencies and updates it via Bayes' rule as new observations are collected. Type-based variants use Dirichlet-multinomial conjugacy for closed-form posterior updates; continuous variants estimate per-information-set action distributions.

[32] Restricted Nash Response (RNR). A safe exploitation algorithm that constructs a strategy blending Nash equilibrium play with best-response exploitation. A safety parameter $p \in [0, 1]$ controls the tradeoff: at $p = 0$ the strategy is a pure best response, and at $p = 1$ it is the Nash equilibrium itself. Formulated as a linear program.

[33] Safe exploitation. The problem of adapting one's strategy to exploit a modeled opponent's weaknesses while maintaining provable guarantees against worst-case loss. In two-player zero-sum games, the Safety Theorem (Ganzfried and Sandholm, 2015) ensures that an exploitation strategy anchored to a Nash equilibrium baseline cannot lose more than the baseline against any opponent.

[34] Teaching attack. An adversarial strategy in which an opponent deliberately plays suboptimally to manipulate an adaptive agent's opponent model, then switches to an exploitative strategy once the agent has been misled. Resilience to teaching attacks is a key robustness criterion for opponent modeling systems.

[35] Adaptation safety. A safety notion (Ge et al., 2024) requiring that an exploitation strategy be no more exploitable than the blueprint baseline strategy from which it was derived. A weaker but more practically achievable guarantee than strict Nash safety, applicable to settings where the baseline is an approximate rather than exact equilibrium.

1.10.5 Multi-Agent Dynamics

[36] Non-stationarity (multi-agent). The fundamental challenge that arises when multiple agents learn simultaneously: each agent's environment — which includes the other agents — changes as those agents update their policies, violating the stationarity assumption of single-agent reinforcement learning.

[37] Centralized Training with Decentralized Execution (CTDE). A multi-agent reinforcement learning paradigm in which agents have access to global information (other agents' observations and actions) during training but execute using only local observations at deployment. Representative algorithms include MADDPG (centralized critic, decentralized actors), QMIX (monotonic value decomposition), and MAPPO (PPO with centralized value function).

[38] Policy Space Response Oracles (PSRO). A population-based framework that maintains a growing set of policies and iteratively computes meta-Nash equilibria over the current population, then trains new best-response policies via reinforcement learning. Unifies fictitious play, self-play, and the double oracle method as special cases.

[39] Population-Based Training (PBT). A meta-optimization framework in which a population of agents trains in parallel, periodically copying weights from high-performing agents (exploit) and mutating hyperparameters (explore). The AlphaStar league is the landmark application, with three agent roles — main agents, main exploiters, and league exploiters — maintaining population diversity and robustness.

[40] Learning with Opponent-Learning Awareness (LOLA). A multi-agent learning algorithm in which each agent differentiates through the anticipated parameter update of its opponent, enabling anticipatory adaptation rather than reactive best-response. Achieves cooperation in settings where naive independent learners converge to suboptimal outcomes.

[41] Coalition. A subset of players in an N-player game who coordinate their strategies for mutual benefit. In free-for-all settings, coalitions may form and dissolve dynamically during play, creating a social structure that cannot be captured by two-player game-theoretic frameworks.

[42] Spinning top decomposition. A decomposition (Balduzzi et al., 2019) of any antisymmetric payoff matrix into a transitive component (representing genuine skill ranking) and a cyclic component (representing non-transitive, rock-paper-scissors structure). The transitive ratio serves as a diagnostic distinguishing real improvement from illusory cycling in population-based training.

[43] Empirical Game-Theoretic Analysis (EGTA). A framework for analyzing multi-agent systems by constructing finite empirical games over sampled strategy sets and computing Nash equilibria of the resulting meta-game. Generalizes two-player exploitability to the population setting and provides approximation bounds for empirical Nash convergence.

[44] Shapley value. A solution concept from cooperative game theory that assigns each player a payoff equal to their average marginal contribution across all possible coalition orderings. Applied in multi-agent reinforcement learning as a credit assignment mechanism (Shapley Q-value) that decomposes joint rewards into per-agent attributions.

[45] So Long Sucker (SLS). A four-player coalition formation game designed by Nash, Shapley, Shubik, and Hausner (1950) in which players must form temporary alliances to eliminate opponents but only one player can win. Serves as a benchmark for studying dynamic coalition formation, betrayal, and negotiation in competitive multi-agent settings.

1.10.6 Data-Driven Approaches

[46] Decision Transformer (DT). A sequence model architecture that reformulates offline reinforcement learning as return-conditioned sequence prediction. A GPT-2 backbone predicts actions from (return-to-go, state, action) token sequences, matching or exceeding value-based offline RL methods without temporal-difference learning or Bellman backups. Subject to a critical limitation in stochastic environments, where return conditioning conflates lucky outcomes with skilled decisions.

[47] Adversarially Robust Decision Transformer (ARDT). An extension of the Decision Transformer that conditions on minimax returns-to-go via expectile regression rather than empirical returns — recovering near-Nash equilibrium strategies from offline data in sequential games with sufficient data coverage. Addresses the standard Decision Transformer's vulnerability in adversarial settings.

[48] Player embedding (player2vec). A vector representation of a player's behavioral patterns, obtained by treating in-game actions as tokens and training a Transformer encoder with masked action prediction on sequences of a player's hands. Produces representations that cluster by behavioral style without requiring manual labels.

[49] VPIP (Voluntarily Put In Pot). A standard poker behavioral statistic measuring the percentage of hands in which a player voluntarily contributes chips to the pot before the flop. Combined with PFR (Pre-Flop Raise percentage), VPIP forms the primary axis for classifying player archetypes (tight versus loose, passive versus aggressive).

[50] Collusion detection. The problem of identifying coordinated play among ostensibly independent players in multiplayer games. Detection signals include co-occurrence anomaly (pairs appearing together more often than expected), chip dumping (directional monetary transfer between colluding players), and soft play (reduced aggression against specific opponents). Extends the coalition detection methodology of Chapter 11 to a practical fraud detection application.

1.10.7 Evaluation and Integration

[51] Approximate exploitability. A method for estimating exploitability in games too large for exact best-response computation, by training an RL-based best-response agent against the strategy under evaluation. The trained agent's payoff provides a lower bound on the true exploitability.

[52] α-Rank. A multi-agent evaluation method based on evolutionary dynamics. Constructs a Markov chain over the strategy population using a Fermi selection function; the stationary distribution assigns each strategy a ranking that captures both transitive dominance and cyclic competitive structure. Unlike Elo, α-Rank is sensitive to intransitive (rock-paper-scissors) relationships.

[53] VasE (Voting as Stochastic Estimation). An evaluation method grounded in social choice theory. Constructs a tournament matrix from pairwise agent comparisons across games and computes maximal lotteries via linear programming. Detects intransitive cycles that Elo-based rankings cannot capture and provides a principled aggregation of performance across heterogeneous evaluation domains.

[54] AIVAT. A variance reduction technique for evaluating agents in imperfect-information games (Burch, Johanson, and Bowling, 2019). Uses a control variate derived from the CFR value function to make counterfactual adjustments at each chance node, reducing evaluation variance by an order of magnitude while maintaining an unbiased estimator.

[55] Marginal exploitability. An extension of standard exploitability to N-player settings. Measures each player's incentive to deviate unilaterally from the current strategy profile, accounting for the fact that in multiplayer games the concept of a unique best response is complicated by coalition structures and non-unique equilibria.